GridWatch: An AI-Powered Energy Poverty Early Warning System

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Abstract

GridWatch is a generative AI system that predicts energy poverty risk at the ZIP-code level before affordability crises occur. By joining DOE LEAD energy burden data, NOAA weather forecasts, EIA electricity price trends, and US Census income and housing data, a machine learning model identifies neighborhoods where energy cost burden is likely to exceed critical thresholds within the next 30–90 days. A large language model layer converts model outputs into human-readable risk explanations for policymakers and assistance programs. This proposal describes the problem, methodology, datasets, feasibility, and a nine-week implementation timeline.

I. INTRODUCTION

Energy poverty — defined as spending more than 10% of household income on energy costs — affects an estimated 30 million US households. Extreme weather events amplify this burden significantly: a single heatwave can cause energy bills to spike 40–60% in low-income neighborhoods with aging housing stock. Government assistance programs such as LIHEAP currently respond reactively, distributing aid after shutoffs occur rather than anticipating where demand will surge.

GridWatch addresses this gap by building a predictive early-warning system that scores ZIP codes by energy poverty risk 30–90 days in advance, allowing proactive resource allocation.

II. PROBLEM STATEMENT

Three compounding factors create the energy poverty prediction gap:

- Reactive aid distribution: LIHEAP and similar programs allocate funds based on prior-year data, missing emerging vulnerable areas.
- Siloed datasets: Energy burden, weather forecasts, and income data exist in separate government databases never joined for predictive modeling.
- No geographic granularity: Existing risk assessments operate at state level, masking ZIP-code and census-tract variation.

III. PROPOSED SOLUTION

GridWatch is a three-layer system. First, a data pipeline ingests and joins four government datasets at the ZIP-code level on a weekly update cycle. Second, an XGBoost classifier predicts which ZIP codes will exceed the 10% energy burden threshold within 30–90 days. Third, a large language model generates plain-language risk narratives from structured model outputs, making predictions actionable for non-technical policymakers.

IV. DATA SOURCES

DOE LEAD (lead.openei.org): Energy burden and low-income energy costs by census tract.

NOAA Climate Data (ncei.noaa.gov): Historical temperature extremes and 30–90 day weather forecasts by region.

EIA State Energy Data (eia.gov): Monthly electricity price trends by state — price spike signal.

US Census ACS (census.gov): Household income, housing age, insulation quality proxies by census tract.

V. METHODOLOGY

(1) Data Integration: Weekly pipeline joins all four sources at ZIP/tract level using GeoPandas spatial joins and Census crosswalk files.

(2) Feature Engineering: Derived features include 30-day temperature anomaly, income-to-energy-cost ratio trend, housing age index, and rolling price change.

(3) Risk Modeling: XGBoost classifier trained on 2018–2023 historical burden data. SHAP values expose which features drive each ZIP-code prediction.

(4) GenAI Narrative Layer: Structured SHAP outputs are passed to an LLM with prompt templates that generate ZIP-code-level risk summaries in plain English.

VI. TIMELINE (MARCH 11 – MAY 4, 2026)

Wk	Milestone
1–2	Data ingestion pipeline; API access confirmed
3	Feature engineering; ZIP-level dataset join
4–5	Risk model training and evaluation (XGBoost)
6	SHAP explainability; GenAI narrative layer
7	Interactive dashboard frontend (React + Mapbox)
8	Testing, tuning, edge-case handling

9	Final polish; showcase preparation

VII. FEASIBILITY

- All four datasets are freely accessible via documented public APIs with Python wrappers.
- XGBoost + SHAP is a well-understood ML stack; training on 5 years of ZIP-level data is computationally lightweight.
- MVP scoped to a single metro area (Los Angeles) to ensure delivery within 9 weeks; national expansion is a stretch goal.
- LLM API costs minimized by caching narratives and batching predictions weekly rather than in real time.

VIII. EXPECTED OUTCOMES

By May 4, 2026: (1) a working web dashboard displaying weekly ZIP-code risk scores for Los Angeles; (2) a trained XGBoost model with validation metrics; (3) an LLM narrative engine generating policy-ready risk summaries; (4) a reproducible pipeline extensible to any US city; and (5) a written report evaluating model performance and policy implications.

IX. CONCLUSION

GridWatch demonstrates that joining four underutilized government datasets with a trained ML model and a generative AI explanation layer can produce a genuinely novel, socially impactful tool that no general-purpose AI assistant can replicate. The project is technically feasible, policy-relevant, and scoped appropriately for a nine-week graduate implementation.

REFERENCES

[1] U.S. Dept. of Energy. "Low-Income Energy Affordability Data (LEAD)." lead.openei.org, 2023.

[2] NOAA National Centers for Environmental Information. ncei.noaa.gov, 2024.

[3] U.S. Energy Information Administration. "State Energy Data System." eia.gov, 2024.

[4] U.S. Census Bureau. "American Community Survey 5-Year Estimates." census.gov, 2022.

[5] Chen, T. and Guestrin, C. "XGBoost: A Scalable Tree Boosting System." KDD, 2016.

[6] Lundberg, S. and Lee, S. "A Unified Approach to Interpreting Model Predictions." NeurIPS, 2017.